

Lightweight Deep Learning Model Architecture for Affective State Recognition from Wearable Physiological Sensors

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Abstract.

In this fast changing world, health of an individual has started taking a huge toll. Industrial leaders are pushing for longer working durations, and people are generally more stressed from all the uncertainty since the post-COVID era. Those who suffered from COVID-19 have also reported weaker lung function, reducing oxygen in the blood and leading to health issues such as depression, anxiety, and stress. In such a phase of life, monitoring health statistics on a regular basis has been a challenge, leading to quite a few avoidable casualties. This has led to the rise of smart bands and smartphones using built-in sensors to monitor the health stats of the user. While these modern devices provide constant monitoring, they fail to provide accurate predictions and preventive insights, thus necessitating the integration of machine learning or deep learning algorithms to analyse these data and provide insightful analysis to help users stay healthy.

Modern implementations for analysing such time-series data include hybrid models of CNN+LSTM architecture which are quite heavy, with parameter counts ranging from around 2M to 6M, longer epoch times during training, and higher activation memory due to larger hidden layers (256/512). The main aim of this research was to make a model much lighter than current state-of-the-art at the cost of minimal performance drop. The architecture proposed consists of CNN+BiGRU. A shallow CNN extracts local temporal patterns from multichannel physiological signals, while a two-layered BiGRU with a hidden size of 128 per direction captures both past and future temporal dependencies within each window — crucial for affective state modelling. The resultant model achieves competitive performance with ~70–85% fewer parameters, demonstrating a favourable efficiency–accuracy trade-off suitable for real-world wearable deployments.

1. Introduction

Designing deep learning models for wearable physiological data requires balancing representational capacity with computational efficiency, particularly for deployment on resource-constrained edge devices. While hybrid CNN–LSTM architectures have become a de facto standard for modelling multivariate physiological time-series due to their ability to capture spatial correlations and long-term temporal dependencies, such models are often characterised by large parameter counts, high memory consumption, and long training and inference times. These limitations prevent practical applicability in real-world wearable health monitoring systems.

To counter the heavy LSTM architecture, BiGRU models have been proposed in this research. GRUs employ a simplified two-gate mechanism compared to the resource-hungry three-gate mechanism in LSTMs. The bidirectional data flow helps the neural network to learn future patterns from past data and vice versa.

This architecture has a shallow convolutional front-end for extracting cross-channel features from multichannel physiological signals, and a two-layer BiGRU for temporal sequence modelling. A multi-head self-attention

mechanism is further employed to selectively emphasise informative temporal segments within each window, enabling the model to focus on critical physiological transitions associated with affective state changes. The resulting representations are aggregated and classified using a compact fully connected head.

This design achieves a favourable efficiency–accuracy trade-off, reducing the overall parameter count by approximately 70–85% compared to conventional CNN–LSTM models while maintaining competitive performance. Such efficiency makes the proposed CNN–BiGRU architecture particularly well suited for continuous stress monitoring and affective state recognition in wearable and edge computing environments.

End-to-End Pipeline Overview

The complete processing pipeline spans six stages, from raw sensor ingestion through final evaluation. The diagram below contextualises where each component of the paper fits within the overall system.

Stage 1 — Raw Data (WESAD .pkl files)	Subjects S2–S17 · Chest RespiBAN ~700 Hz · Wrist Empatica E4 ~32 Hz · 4 affective labels
Stage 2 — Preprocessing	Upsample wrist · Compute ACC magnitude · Downsample to 32 Hz · Stack 8 channels · Sliding window 60 s / 50%
Stage 3 — Exploratory Data Analysis	Class distribution · Per-subject heatmap · UMAP / t-SNE projection · Raw signal snapshot
Stage 4 — Model (CNN -> BiGRU -> LayerNorm -> Multi-Head Attention -> Classifier)	Input (B, 1920, 8) -> ~1.2 M parameters -> Output (B, 4) logits
Stage 5 — LOSO Training	AdamW (lr = 0.001) · WeightedRandomSampler · EarlyStopping (patience = 6) · AMP · GradClip (max = 5.0)
Stage 6 — Evaluation & Comparison	Accuracy · Macro-F1 · Confusion matrix · Baseline vs. Balanced · SOTA comparison

Figure 1. End-to-end pipeline — six stages from raw WESAD recordings to LOSO evaluation.

2. Literature Survey

Wearable devices in the healthcare industry have become a significant development, offering constant monitoring. For better patient outcomes, the monitored data should be analysed on-device with the support of technologies such as blockchain, machine learning, deep learning, and IoT. The following summarises key contributions made by other researchers in this direction.

[1] Alattar and Mohsen conducted a survey on the use of smart wearable devices in healthcare, examining how devices are powered, how they connect with other devices, and how comfortable they are for patients. Although the study helps in selecting hardware for the general population, it lacks in-depth clinical testing to confirm device efficiency.

[2] Xie, Li, and Liu introduced SHARP: a blockchain-powered wireless sensor network for real-time student health monitoring and personalised learning. Using a deep neural network (DNN), the system classifies health states such as fatigue and stress, achieving a 94.2% F1-score in simulated classroom environments. However, dependence on simulation raises concerns about real-world scalability and acceptance across geographies.

[3] Ribeiro et al. carried out a systematic review on continuous movement monitoring at home through wearable devices. These systems support chronic disease management and promote personalised healthcare. While they reduce hospital visits and improve patient management, challenges persist in data privacy, device usability, and

patient compliance.

[4] Kumar and Trinh explored advances in wearable biosensors for healthcare, focusing on analytics using simulations in Proteus Professional Software. However, scalability and reliability in real environments remain untested.

[5] Vijayan et al. reviewed wearable IoT devices for remote health monitoring, integrating real-time physiological data with cloud-based machine learning models to detect anomalies. Although practical, issues of real-time data processing and AI interpretability were identified.

[6] Khan et al. proposed an AI-driven real-time health monitoring system using wearable sensors. Physiological signals such as blood glucose, SpO_2 , and heart rate were analysed using traditional ML models (Random Forest, SVM) and deep learning (CNN, LSTM), demonstrating good early disease prediction. The main challenge was the requirement for broader population-level validation.

[7] Mahabub et al. reviewed wearable technologies for health monitoring, including photoplethysmography (PPG) and galvanic skin response (GSR). Future trends towards multi-modal inputs and AI-driven interventions were discussed, though the work relies on secondary data without experimental validation.

[8] Nongrum et al. reviewed challenges of collecting data from mobile devices in connected health systems, highlighting data interoperability, heterogeneous sensor formats, and large-scale real-time integration. This underscores the importance of robust infrastructure alongside wearable hardware.

[9] Mhatre et al. examined computational approaches for the analysis of wearable health data and early disease detection using CNNs and RNNs on continuous physiological data streams. Challenges including noisy sensor data, inconsistent sampling rates, and privacy concerns were emphasised.

[10] Kristoffersson and Lindén conducted a systematic review of wearable sensors for monitoring physical activity, noting challenges in correlating sensor biomarkers with clinical gold standards and maintaining sensor stability under varying conditions.

Recent studies have shifted from traditional feature-based machine learning to end-to-end deep learning models due to their robustness in handling anomalies. Hybrid CNN-LSTM architectures have been widely adopted for their ability to capture spatial features and long-term temporal dependencies. The main trade-off is that these architectures are resource-hungry post-deployment, contributing to high computational cost during training, evaluation, and deployment — highlighting the need for lightweight and efficient deep learning architectures.

3. Research Methodology

3.1 Data Preprocessing

The WESAD dataset comprises physiological signals collected from chest-worn (RespiBAN) and wrist-worn (Empatica E4) sensors across 15 subjects covering 4 affective states: baseline, stress, amusement, and meditation. The pipeline addresses multi-modal sensor fusion and temporal alignment challenges inherent in heterogeneous data.

Raw data for each subject were extracted from pickle files containing chest sensors sampled at approximately 700 Hz (ECG, EDA, respiration, temperature, and 3-axis accelerometer) and wrist sensors sampled at approximately 32 Hz (EDA, temperature, and 3-axis accelerometer). The wrist and chest signals were captured at different frequencies, leading to discrepancies in the length of information gathered per unit of time. This issue was addressed by upsampling wrist signals to match chest signals using linear interpolation.

Feature engineering was applied to accelerometer data from both sensors. Three-axis signals (x , y , z) were transformed into scalar magnitude features using the Euclidean norm ($\sqrt{x^2 + y^2 + z^2}$). This reduced

dimensionality from three features per accelerometer to one while preserving movement intensity information.

All physiological signals were then downsampled to a uniform sampling rate of 32 Hz using block-based averaging, which preserves signal characteristics, reduces high-frequency noise, and significantly reduces computational requirements. A sampling rate of 32 Hz is sufficiently faithful to record physiological dynamics (heart rate variability, breathing patterns, electrodermal responses) while keeping model training times practical.

The eight physiological channels were transformed into a single multi-modal feature matrix, providing the model with a comprehensive picture of the body's response to stress across various physiological domains.

A sliding window segmentation method was applied to generate fixed-length temporal sequences compatible with deep learning. Windows of 60 s were used with 50% overlap, producing signal segments of 1920 time steps (32 Hz $\times$ 60 s) across 8 channels. The 50% overlap augments the number of training samples and ensures smooth coverage of temporal transitions between affective states.

A 70% dominance threshold was used to assign window-level labels. A label was assigned to a window only when the most frequent label constituted at least 70% of the window's samples; windows not meeting this threshold were classified as baseline. This approach ensures only unambiguous segments are used for training, reducing label noise that may hurt classifier performance.

3.2 Model Architecture

The model architecture comprises a Convolutional Neural Network (CNN), Bidirectional Gated Recurrent Units (BiGRU), and multi-head self-attention mechanisms. It processes pre-processed 60 s windows through five sequential stages: CNN-based feature extraction, bidirectional recurrent processing, layer normalisation, multi-head attention, and classification.

Stage	Shape	Details
Input	(B, 1920, 8)	(B, 1920, 8) — 8 sensor channels $\times$ 1920 timesteps (60 s @ 32 Hz)
CNN Front-end	(B, 1920, 64)	Conv1D(8 $\rightarrow$ 64, k=3) $\cdot$ ReLU $\cdot$ Conv1D(64 $\rightarrow$ 64, k=3) $\cdot$ ReLU $\cdot$ Dropout(0.2)
BiGRU (2 layers)	(B, 1920, 256)	Bidirectional GRU hidden=128 $\cdot$ concat $\rightarrow$ (B, 1920, 256) $\cdot$ Dropout(0.3) inter-layer
LayerNorm + Multi-Head Attn	(B, 256)	LayerNorm(256) $\cdot$ 4-head self-attention $\cdot$ Mean pool $\rightarrow$ (B, 256)
Classifier Head	(B, 4)	Linear(256 $\rightarrow$ 128) $\cdot$ ReLU $\cdot$ Dropout $\cdot$ Linear(128 $\rightarrow$ 4) — Kaiming/Xavier init
Output	(B, 4)	(B, 4) logits $\rightarrow$ CrossEntropyLoss $\rightarrow$ {Baseline, Stress, Amusement, Meditation}

Figure 2. Model block diagram — tensor shapes at each stage. B = batch size (128), T = 1920 timesteps.

3.2.1 CNN Front-End — Local Feature Extraction

Two sequential 1D convolutional layers with 64 filters and kernel size 3 extract spatial features, acquiring inter-channel interaction patterns. ReLU activations introduce non-linearity while Dropout(0.2) regularises the network. The CNN transforms raw multi-channel signals into 64 learned feature representations capturing physiological correlations.

The key design choice is same-padding (pad=1): each output timestep sees itself and one neighbour on each side (~ 93 ms at 32 Hz), detecting short local events such as ECG peaks and EDA spikes without shrinking the sequence. Kaiming uniform initialisation is applied to convolutional layers, appropriate for ReLU activations.

Layer	Parameters	Operation
Conv1D Layer 1	in=8, out=64, kernel=3, pad=1	Same-padding -> T preserved · ReLU
Conv1D Layer 2	in=64, out=64, kernel=3, pad=1	Refines features · ReLU · Dropout(0.2)

 Figure 3. CNN front-end — same-padding (pad=1) preserves $T=1920$ through both convolutional layers.

3.2.2 Bidirectional GRU — Temporal Context

The two-layered BiGRU with 128 hidden units processes sequences in both forward and backward temporal directions and concatenates the results to generate 256-dimensional representations at each time step. The forward pass records gradual physiological variations whereas the backward pass captures contextual patterns and anticipatory responses. Inter-layer Dropout(0.3) prevents overfitting due to the small subject pool, and Layer Normalisation stabilises training by normalising activation scales.

Gate / State	Equation	Role
Reset gate r_t	$r_t = \text{sigma}(W_r x_t + U_r h_{t-1} + b_r)$	Controls how much past hidden state to expose to the candidate
Update gate z_t	$z_t = \text{sigma}(W_z x_t + U_z h_{t-1} + b_z)$	Interpolation knob: old memory vs. new candidate (0=keep, 1=replace)
Candidate $\tilde{h}_t$	$\tilde{h}_t = \text{tanh}(W_h x_t + U_h (r_t \circ h_{t-1}) + b_h)$	Proposed new hidden state; r_t gates past memory exposure
Final hidden h_t	$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t$	Learned blend; $z \sim 0$ keeps past (slow signals), $z \sim 1$ replaces (state transitions)
BiGRU concat	$[h_{\text{fwd}} \parallel h_{\text{bwd}}]$ at each timestep	128d forward + 128d backward = 256d per timestep; captures both anticipatory and lagging physiology

Table 1. GRU gate equations and their physiological interpretation.

Key insight on the update gate: when $z_t \sim 0$ the model retains old hidden state unchanged — allowing slow physiological signals like skin temperature to persist across many timesteps. When $z_t \sim 1$ the model replaces memory with the new candidate — appropriate during sharp affective state transitions such as baseline to stress.

3.2.3 Multi-Head Self-Attention — Global Re-Weighting

The self-attention mechanism computes relevance scores between all time-step pairs, enabling the model to emphasise critical moments — such as stress peaks or sudden physiological transitions — while de-emphasising baseline periods. Multiple attention heads learn complementary importance patterns, allowing the model to extract discriminative features from heterogeneous 60 s windows where stress indicators may appear at different temporal locations.

Each head operates in a 64-dimensional subspace ($256/4 = 64$ per head). Queries Q, Keys K, and Values V are obtained by separate learned projections of the BiGRU output. Division by $\text{sqrt}(d_k) = \text{sqrt}(64) = 8$ prevents dot products from growing too large, keeping softmax gradients from vanishing. The four head outputs are concatenated and projected back to 256 dimensions via W^O .

Step	Shape Transform	Description
Q, K, V projection	$(B, 1920, 256) \rightarrow (B, 1920, 64)$	Separate learned projections W_q, W_k, W_v per head
Scaled dot-product	$(B, 1920, 64) \rightarrow (B, 1920, 1920)$	scores = $QK^T / \text{sqrt}(64) \rightarrow \text{softmax} \rightarrow \alpha$

Step	Shape Transform	Description
Attention output	(B,1920,1920) -> (B,1920,64)	$\alpha \cdot V$ per head
Concat + W^O	(B,1920,256) -> (B,1920,256)	4 heads concatenated · linear projection W^O
Mean pool	(B,1920,256) -> (B,256)	Mean over $T=1920$ timesteps; parameter-free

Figure 5. Query-Key-Value projection and scaled dot-product attention flow across 4 parallel heads.

3.2.4 Temporal Pooling and Classification Head

Temporal mean pooling aggregates the attention-weighted sequence into a fixed-length representation by averaging across time steps. The classifier head consists of two fully connected layers with ReLU activation and dropout. Kaiming initialisation is used for convolutional and linear layers; Xavier initialisation is used for GRU weights, following best practices for their respective activation functions.

3.3 Training and Evaluation

Leave-One-Subject-Out (LOSO) cross-validation was used for training and evaluation. In each fold, one subject was held out as the test set while the remaining subjects were used for training. This procedure is repeated with each subject serving as the test subject exactly once across all 15 folds.

3.3.1 Normalisation and Class Balancing

Data were normalised using z-score standardisation. Channel-wise means and standard deviations were computed via streaming computation to maintain memory efficiency, then applied to normalise all signals to zero mean and unit variance. This fold-specific normalisation prevents information leakage from test subjects while ensuring consistent input distributions across channels with inherently different scales.

The WESAD dataset has class imbalance, with most data points belonging to the baseline class. This was addressed through weighted random sampling during training. Sample weights were computed as the inverse of class frequencies, causing minority classes to be oversampled in each training batch. This strategy ensured the model received approximately equal exposure to all affective states during training, preventing bias towards the majority baseline class. The WeightedRandomSampler with replacement generated balanced mini-batches of 128 samples throughout training.

3.3.2 Optimisation and Regularisation

AdamW optimiser was used for model training with an initial learning rate of 0.001 and cross-entropy loss for classification. Mixed precision training (AMP) was used to maximise GPU usage and accelerate training time, ensuring numerical stability through automatic loss scaling. Gradient clipping with a maximum norm of 5.0 prevented training instability from exploding gradients, particularly important given the recurrent BiGRU layers. Training proceeded for a maximum of 25 epochs with early stopping based on validation loss, terminating if no improvement occurred for 6 consecutive epochs. The model checkpoint with the lowest validation loss is retained for evaluation.

The test set is evaluated using: overall accuracy, macro-averaged F1 score, per-class precision, recall and F1 scores, and confusion matrices. Comparative analysis evaluated the impact of balanced sampling by contrasting results with and without the weighted sampler, quantifying improvements in minority class recognition.

Step	Formula	Purpose
Gradient	$g_t = \text{grad } L(\theta_t)$	Backpropagation through CNN to BiGRU to Attn to Head

Step	Formula	Purpose
1st moment m_t	$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$, $\beta_1=0.9$	Exponential moving average — smooths noisy recurrent gradients
2nd moment v_t	$v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$, $\beta_2=0.999$	Moving average of squared gradient — per-parameter adaptive learning rate
Bias correction	$m^\wedge = m_t / (1-\beta_1^t)$, $v^\wedge = v_t / (1-\beta_2^t)$	Corrects initialisation bias; critical in early training when moments start at 0
Param update	$\theta \leftarrow \alpha \cdot m^\wedge / (\sqrt{v^\wedge} + \epsilon)$, $\alpha=0.001$	Adaptive per-param lr: large step for small-grad params (attention), small for large-grad (GRU)
Weight decay (decoupled)	$\theta \leftarrow \lambda \cdot \theta$, $\lambda=0.01$	Applied after gradient step — decoupling prevents adaptive scaling from undermining regularisation
Gradient clipping	if $\ g\ > 5.0$: $g = g \cdot 5.0/\ g\ $	Rescales direction preserved; prevents BiGRU exploding gradients at affective state boundaries

Table 2. AdamW optimiser — step-by-step equations and rationale.

3.3.3 Forward and Backward Pass

Each training step involves a complete forward pass through all model components, loss computation, and a backward pass that propagates gradients in reverse. PyTorch's autograd engine builds a dynamic computation graph during the forward pass and traverses it in reverse during backward. The AMP GradScaler runs the forward pass in FP16 for speed, then unscales gradients to FP32 before clipping and the AdamW parameter update.

Forward Step	Forward Detail	Backward Step	Backward Detail
DataLoader	$X (128,1920,8) \cdot y (128,)$ labels	AdamW.step()	Update all θ using $m^\wedge$, $v^\wedge$, weight decay $\lambda=0.01$
CNN	$(128,1920,64) \cdot$ stores conv activations	Clip grad	$\ g\ > 5 \rightarrow$ rescale to radius 5, direction kept
BiGRU	$(128,1920,256) \cdot$ stores all hidden states	unscale_()	AMP: restore FP32 grads before clipping
Attention + Pool	$(128,256) \cdot$ stores alpha weights	backward()	AMP scaled loss $\cdot$ autograd chain rule
Classifier	$(128,4)$ logits $\cdot$ stores pre-activations	zero_grad()	Clear grads $\cdot$ set_to_none=True
CE Loss	scalar $\cdot$ AMP scaled for FP16 backprop	dL/d(logits)	$\text{softmax}(z) - \text{one_hot}(y) \cdot$ seed gradient

Figure 7. Complete forward and backward pass for a single training step.

4. Mathematical Foundations of the Architecture

This section presents the mathematical formulations underlying each component of the proposed CNN–BiGRU architecture, from signal preprocessing through classification and evaluation. Equations are grouped by pipeline stage.

Eq.	Formula / Expression	Description & Variables
(1)	$\text{magnitude} = \sqrt{x^2 + y^2 + z^2}$	ACC Magnitude — collapses 3-axis data to scalar, eliminates orientation bias, applied to both chest and wrist sensors
(2)	$f(t_{\text{new}}) = f(t_i) + [(t_{\text{new}} - t_i) / (t_{i+1} - t_i)] \cdot (f(t_{i+1}) - f(t_i))$	Linear Interpolation (Wrist Upsampling) — upsamples wrist ~32 Hz to chest ~700 Hz; grids normalised to [0,1]

Eq.	Formula / Expression	Description & Variables
(3)	$x_d[k] = (1/D) \sum_{i=0}^{D-1} x[k \cdot D + i]$	Block-Based Averaging (Downsample to 32 Hz) — k =block index, $D=f_{src}/f_{tgt}$; preserves signal energy
(4)	$label(W) = \text{argmax}_c \text{count}(c, W) \text{ if } \text{count}/N \geq 0.70$	Majority Label Threshold — window labelled only when dominant class covers $\geq 70\%$ of $N=1920$ samples; reduces transition noise
(5)	$\mu_c = (1/N) \sum x_i^c$; $\sigma_c = \sqrt{\text{max}(\text{var}, \text{eps})}$; $x^\wedge = (x - \mu)/\sigma$	Z-Score Standardisation (streaming, channel-wise, fold-specific) — $\text{eps}=10^{-12}$; computed over training subjects only
(6)	$w_c = 1/\text{max}(\text{count}_c, 1)$; $\text{sample_weight}_i = w_{c_i}$	Inverse-Frequency Weighting — WeightedRandomSampler draws 128-sample balanced batches with replacement
(7)	$W \sim \text{KaimingUniform}(a=\sqrt{5})$; $W_{GRU} \sim \text{XavierUniform}$	Weight Initialisation — Kaiming for Conv1D/Linear (ReLU); Xavier for GRU recurrent weights (tanh/sigmoid gates)
(8)	$LN(x) = \text{gamma} \cdot (x - \mu) / \sqrt{\text{sigma}^2 + \text{eps}} + \text{beta}$	Layer Normalisation — applied to BiGRU output ($d=256$) per timestep; gamma, beta learnable; eps stabilises variance
(9a)	$\text{Attn}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$	Scaled Dot-Product Attention — $d_k=64$ ($256/4$ heads); $1/\sqrt{d_k}$ prevents softmax saturation
(9b)	$\text{MultiHead} = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \cdot W^O$	Multi-Head Attention — $H=4$ heads; W^O projects $\text{concat}(256)$ back to $\text{embed_dim}(256)$
(10)	$\text{pooled} = (1/T) \sum_{t=1}^T \text{attn_out}[t]$	Temporal Mean Pooling — aggregates $T=1920$ vectors (each in $\mathbb{R}^{256}$) to fixed $(B, 256)$; parameter-free
(11)	$L = -\sum_c y_c \cdot \log(\text{softmax}(z_c))$; $p_c^\wedge = \exp(z_c) / \sum_k \exp(z_k)$	Cross-Entropy Loss — $C=4$ classes; y_c in $\{0,1\}$ one-hot; z_c =logit; batch loss = mean over 128 samples
(12)	$\ \text{grad}\ _2 = \sqrt{\sum g_i^2}$; $\text{grad}' = \text{grad} \cdot \min(1, \text{theta}_{\text{max}} / \ \text{grad}\ _2)$	Gradient Clipping (L2 Norm) — $\text{theta}_{\text{max}}=5.0$; rescales uniformly when norm exceeds threshold
(13)	$\text{CM}_{\text{norm}}[i,j] = \text{CM}[i,j] / \sum_k \text{CM}[i,k]$	Confusion Matrix Row Normalisation — aggregated across 15 LOSO folds then row-normalised; diagonal = per-class recall
(14)	$\text{Acc} = \text{correct}/\text{total}$; $P_c = \text{TP}/(\text{TP}+\text{FP})$; $R_c = \text{TP}/(\text{TP}+\text{FN})$; $F1_c = 2PR/(P+R)$; $\text{Macro-F1} = (1/C) \sum F1_c$	Accuracy, Precision, Recall, F1 & Macro-F1 — Macro-F1 weights all $C=4$ classes equally; reported as mean $\pm$ std over 15 folds

Table 3. Complete mathematical reference — preprocessing, architecture, training, and evaluation.

5. Results

Exploratory data analysis revealed substantial class imbalance with baseline windows comprising ~ 65 – 75% of the data, with the remaining affective states comprising the rest. UMAP dimensionality reduction showcased moderate class separability with four distinct clusters and partial overlap between meditation and amusement.

5.1 Baseline Training Results

Baseline training achieved a mean accuracy of 0.62 ± 0.09 and macro-F1 of 0.27 ± 0.11 across LOSO folds, with per-subject accuracy ranging from 0.45 to 0.77. The normalised confusion matrix shows correct identification of 84% of baseline windows, but only 25% of stress, 0% of amusement, and 21% of meditation instances. Per-class F1 scores quantified this imbalance: baseline achieved 0.78, while stress (0.18), amusement (0.03), and meditation (0.18) remained near chance performance.

5.2 Balanced Training Results

Implementing weighted random sampling to address class imbalance yielded mixed results. Mean accuracy remained comparable at 0.64 ± 0.09 , while macro-F1 showed marginal improvement to 0.28 ± 0.09 . The balanced approach substantially improved stress detection, increasing its F1 score from 0.18 to 0.26 (+44% relative improvement) and recall from 25% to 34%. Baseline class performance remained stable (F1 = 0.78), demonstrating that oversampling minorities did not degrade majority class recognition.

However, amusement detection showed no meaningful improvement (F1 remained near 0.00), with 92% of instances still misclassified as baseline. Meditation performance slightly declined from F1 of 0.18 to 0.13, suggesting potential negative transfer from increased exposure to other minority classes during training.

Metric	Baseline (no sampler)	Balanced (WeightedSampler)	Change
Mean Accuracy	0.62 ± 0.09	0.64 ± 0.09	+0.02
Macro-F1	0.27 ± 0.11	0.28 ± 0.09	+0.01
F1 — Baseline	0.78	0.78	—
F1 — Stress	0.18	0.26	+0.08 (+44%)
F1 — Amusement	0.03	~0.00	No change
F1 — Meditation	0.18	0.13	−0.05
Stress recall	25%	34%	+9 pp
Amusement recall	0%	1%	No change

Table 4. Results summary — baseline vs. balanced sampling across LOSO folds.

6. Comparison with State-of-the-Art Methods on WESAD

Table 5 presents a comparison of the proposed CNN–BiGRU model against published methods evaluated on the WESAD dataset. A critical distinction must be noted: studies using subject-specific (personalised) validation routinely report accuracies of 95–99%, while subject-independent LOSO evaluation yields substantially lower but more clinically realistic estimates. The proposed model is evaluated exclusively under strict LOSO cross-validation, making it directly comparable only to other LOSO-based methods. Furthermore, most prior work addresses binary (stress vs. non-stress) or 3-class problems; this work tackles the harder 4-class (baseline, stress, amusement, meditation) classification task.

Study	Model	Classes	Eval.	Acc. (%)	Macro-F1	Params
Schmidt et al. [11] (2018)	AdaBoost / LDA	3	LOSO	80.34	74.43	N/A
Bobade & Vani [12] (2020)	DNN	3	LOSO	84.32	N/A	N/A
Melchiades et al. [13] (2022)	RNN	3	LOSO	86.00	N/A	N/A
Gil-Martin et al. [14] (2022)	CNN + FFT	Multi	LOSO	N/R	N/R	Heavy
Sah et al. [15] (2021)	CNN	Multi	Subject-sp.	92.85	N/A	N/R
Ghosh et al. [16] (2022)	GAF + DNN	4	Subject-sp.	94.80	N/A	Heavy
Bayat et al. [17] (2021)	CNN + Transformer	2	LOSO	N/R*	N/R	Heavy
Bansal et al. [18] (2025)	StressNet-GAF + VGG-16	3	Subject-sp.	91.00	N/A	138M+

Study	Model	Classes	Eval.	Acc. (%)	Macro-F1	Params
PAMG-AT [19] (2026)	GNN + Attn + Transformer	2	LOSO	94.59	N/R	Heavy
Proposed CNN-BiGRU (Ours)	CNN + BiGRU + Attn	4	LOSO	64.00	28.00	~1.2 M

Table 5. Comparison of deep learning methods on the WESAD dataset.

N/A = metric not applicable or not reported for that task; N/R = metric not reported by authors; N/R = accuracy reported only with test-set calibration; Subject-sp. = subject-specific (personalised) validation, not directly comparable to LOSO; Heavy = model parameter count not reported but architecture known to exceed 6M+ parameters; Multi = multi-class (3+). Proposed model is the only method evaluated on strict 4-class LOSO with a sub-2M parameter budget.*

7. Discussion

The results demonstrate that the CNN+BiGRU architecture successfully captures physiological patterns for affective state classification, achieving strong baseline class performance.

7.1 Efficiency vs. Capacity Trade-off

The GRU's simplified gating mechanism (only reset and update gates) compared to LSTM (input, forget, and output gates) reduces parameter count by 25–30% for equivalent hidden dimensions. For the implemented configuration with 128 hidden units across two layers, the BiGRU comprises ~1.2M parameters compared to 1.6–2M for an equivalent BiLSTM, yielding a lighter model without loss of architectural depth or capability.

Experimental observations showed a 35–40% reduction in training time compared to a BiLSTM architecture. Memory footprint during inference decreases proportionally, reducing GPU memory consumption by 20–25% — a crucial factor for deployment on edge devices with limited resources such as mobile processors. This efficiency-performance trade-off makes CNN+BiGRU particularly suitable for real-world affective state monitoring.

7.2 Comparative Analysis with Prior Work

Interpreting the results in Table 5 requires careful attention to evaluation methodology. The majority of prior WESAD studies report high accuracy figures (91–95%) under subject-specific or personalised validation, where each model is tuned on a fraction of the test subject's own data. Such protocols are not representative of real-world deployment where a pre-trained model must generalise to entirely unseen individuals. The proposed CNN-BiGRU model is evaluated exclusively under strict LOSO cross-validation across all 15 subjects, providing a far more conservative and clinically meaningful estimate of generalisation capability.

The original WESAD paper by Schmidt et al. [11] established the reference LOSO baseline for 3-class classification at 80.34% accuracy (F1: 74.43%) using conventional classifiers such as AdaBoost and LDA on hand-crafted features. Bobade and Vani [12] improved upon this using deep neural networks, achieving 84.32% accuracy under LOSO for a 3-class task. Melchiades et al. [13] reported 86% accuracy with an RNN under LOSO. These three studies represent the most directly comparable benchmarks — all use LOSO validation, though they address 3-class rather than 4-class problems. The proposed model achieves 64% accuracy and macro-F1 of 0.28 on the harder 4-class task under identical LOSO protocol, which is lower in absolute terms but reflects the added difficulty of distinguishing amusement and meditation — states that are physiologically similar and highly underrepresented in the dataset.

Gil-Martin et al. [14] proposed a CNN combined with Fourier Transform preprocessing under LOSO on WESAD, targeting a multi-class scenario. Their architecture does not report explicit parameter counts but relies on heavier convolutional stacks unsuitable for edge deployment. Similarly, Bayat et al. [17] combined convolutional layers with a Transformer for ECG-based binary stress detection under LOSO, reporting competitive results only when fine-tuned with test-subject data — a semi-personalised setting. In contrast, the proposed model requires no test-time calibration and operates in fully subject-independent mode.

Heavier models leveraging image-encoding strategies — notably Ghosh et al. [16] with Gramian Angular Field (GAF) encoding achieving 94.8% (subject-specific, 4-class) and Bansal et al. [18] with StressNet-GAF and VGG-16 achieving 91% (subject-specific, 3-class) — demonstrate that encoding time-series as 2D images before applying pretrained convolutional networks is powerful. However, VGG-16 carries approximately 138M parameters, rendering such models entirely impractical for edge wearable deployment. The PAMG-AT model [19], the most recent LOSO-based comparison, achieves 94.59% for binary stress classification using a hierarchical graph neural network with multi-level attention and Transformer components. While impressive, it targets a 2-class problem and employs an architecture of considerable complexity unlikely to be deployable on microcontroller-grade hardware.

The key differentiating contribution of the proposed work therefore lies not in absolute accuracy — which is naturally bounded by the hardness of the 4-class LOSO setting and severe class imbalance — but in the model's computational profile. At approximately 1.2M parameters, the proposed CNN-BiGRU is 70–85% lighter than equivalent CNN-LSTM architectures (4M to 8M parameters) and requires 35–40% less training time and 20–25% less GPU memory during inference. No prior WESAD study simultaneously achieves LOSO-validated 4-class classification within a sub-2M parameter budget.

8. Conclusion

This paper introduces a lightweight, high-performance deep learning architecture for affective state recognition from physiological wearable sensors, addressing a critical gap between model performance and deployability on resource-constrained systems. The proposed architecture — combining a shallow CNN for spatial feature extraction with a two-layer BiGRU and self-attention mechanism — effectively captures both local and long-range temporal dependencies while significantly reducing parameter count and computational cost compared to standard CNN-LSTM architectures.

Experimental evaluation on the WESAD dataset under leave-one-subject-out cross-validation demonstrates that the model attains competitive accuracy without high training time or memory requirements, making it highly applicable to real-life continuous stress monitoring. Class imbalance remains a challenge, especially for minority affective states, but the results prove that a good accuracy–efficiency trade-off is achievable. The proposed CNN-BiGRU model therefore represents a plausible solution for practical wearable-based affective computing.

9. Future Scope

The model's performance is fundamentally constrained by severe class imbalance in the WESAD dataset. The ability to achieve 0.64 overall accuracy despite this imbalance demonstrates the architecture's strength.

An important direction for future work involves evaluating the architecture on datasets with more balanced class distributions or larger minority class samples. Given the model's computational efficacy (~1.2M parameters, 40% faster training than LSTM alternatives), this architecture is promising for real-world stress monitoring systems once dataset imbalance has been addressed.

Appendix A — Architectural Design Decisions

This appendix documents the rationale behind each key architectural and training hyperparameter choice. These decisions collectively define the efficiency–accuracy trade-off achieved by the proposed model.

Design Choice	Value / Setting	Rationale
CNN kernel_size	3, same-padding	Receptive field ~93 ms at 32 Hz — tight enough for ECG peaks and EDA spikes without blurring slow channels (temperature)
BiGRU hidden size	128 per direction (256 total)	Large enough for complex temporal patterns; 25–30% fewer parameters than equivalent BiLSTM with same hidden size
GRU over LSTM	2-gate vs. 3-gate mechanism	GRU eliminates the separate cell state and output gate of LSTM, reducing parameters by ~25–30% for equivalent capacity
2-layer BiGRU	2 stacked layers	Adds representational depth; inter-layer Dropout(0.3) provides regularisation given the small subject pool (15 subjects)
4-head attention	$d_k = 64$ per head	$256/4=64$ ensures divisibility; 4 heads enable specialisation across local/global/cross-modal attention patterns
Mean pooling	over $T=1920$	More robust than CLS token or last hidden state for variable-length physiological windows with class label anywhere in window
Window size	60 s / 1920 timesteps	Captures full slow-timescale physiology (cortisol-linked EDA, breathing cycles); 50% overlap augments training samples
Majority label threshold	70% dominance	Stricter than 60% — ensures only unambiguous affective state windows are used for training, reducing boundary label noise
WeightedRandomSampler	Inverse class frequency	Preferred over loss weighting — affects what the model sees during training rather than only how gradients are scaled
AdamW	$lr=0.001$, $wd=0.01$, decoupled weight decay	Decoupling prevents adaptive scaling from undermining L2 regularisation for infrequently-updated parameters
Grad clip = 5.0	Global L2 norm threshold	Conservative for 2-layer BiGRU; gradient norms can spike sharply at affective state transitions in 60 s windows
AMP (FP16)	GradScaler + <code>unscale_()</code>	Reduces GPU memory ~20–25% and accelerates training; <code>unscale_()</code> called before clipping to restore FP32 gradient magnitudes

Table A1. Architectural and training design decisions with rationale.

Appendix B — LOSO-Specific Failure Modes and Mitigations

Leave-One-Subject-Out cross-validation introduces failure modes distinct from standard overfitting. The four risks below are specific to cross-subject physiological evaluation and are each explicitly addressed in the pipeline design.

Risk	Description	Mitigation in Pipeline
Subject distribution shift	Held-out subject may have physiological baseline (EDA amplitude, HR range) entirely outside the training distribution	Z-score normalisation computed per fold from training subjects only
Label imbalance per fold	Removing one subject shifts per-class frequency; minority classes become even rarer in some folds	WeightedRandomSampler recomputed per fold from training class frequencies

Risk	Description	Mitigation in Pipeline
Normalisation leakage	Including test subject statistics in normalisation gives the model indirect knowledge of the test distribution	Streaming mean/std computation strictly over training files; test never accessed until evaluation
Window boundary artefacts	Windows near label transitions contain mixed-state physiological signal, leading to noisy labels	70% dominance threshold for window-level label assignment; borderline windows classified as baseline

Table B1. LOSO-specific failure modes and their mitigations.

Appendix C — Complete Tensor Shape Reference

The table below provides the complete input/output shape and key parameters for every layer in the model. B = batch size (128 during training); T = 1920 (timesteps per window = 60 s × 32 Hz). Attention shapes assume single-head view; 4 heads run in parallel.

Layer	Input	Output	Key Parameters
CNN Layer 1	(B, 1920, 8)	(B, 1920, 64)	kernel=3, pad=1, stride=1, ReLU
CNN Layer 2	(B, 1920, 64)	(B, 1920, 64)	kernel=3, pad=1, ReLU, Dropout(0.2)
BiGRU	(B, 1920, 64)	(B, 1920, 256)	hidden=128, layers=2, bidirectional, Dropout(0.3) inter-layer
LayerNorm	(B, 1920, 256)	(B, 1920, 256)	Normalise over dim=256 per timestep per sample
Q/K/V projection	(B, 1920, 256)	(B, 1920, 64)	W_q, W_k, W_v in $\mathbb{R}^{256 \times 64}$, one set per head
Attn scores	(B, 1920, 64)	(B, 1920, 1920)	$QK^T / \sqrt{64}$, softmax over last dim
Attn output	(B, 1920, 1920)	(B, 1920, 64)	$\alpha \cdot V$ per head
Concat + W^O	(B, 1920, 256)	(B, 1920, 256)	4 heads concatenated (256d) + linear projection W^O
Mean pool	(B, 1920, 256)	(B, 256)	Mean over T=1920; parameter-free
Linear 1	(B, 256)	(B, 128)	ReLU, Dropout(0.3), Kaiming init
Linear 2	(B, 128)	(B, 4)	4-class logits, Kaiming init, no activation

Table C1. Complete layer-by-layer tensor shape reference.

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